Charles Fucile

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Education

Georgia Institute of Technology Aug 2024 – May 2028
Bachelor of Science in Aerospace Engineering – *John H. Martinson Honors Program* GPA: **3.80**
Relevant Coursework: Aerodynamics, Computational Fluid Dynamics, Thermodynamics, Jet & Rocket Propulsion, Structural Analysis, Deformable Bodies, Vehicle Performance, System Dynamics, Experimental Methods

Technical Experience

Aircraft Window Proof Fixture Modernization Intern Aug 2026 – Dec 2026
NASA Armstrong Flight Research Center

- Developed a new **pressure vessel** proof test fixture for **8 distinct aircraft** using **Creo modeling**, **GD&T**, and **piping schematic design**, improving test process efficiency by up to **50%** relative to the previous configuration.
- Executed **FEA and hand calculations** for structural analysis of the fixture design, reducing **stored energy by over 80%** for improved operator safety between design versions.
- Took full ownership of design, drawing, and fabrication management for **20+ mechanical hardware designs** supporting airborne science mission operations, serving as the primary designer across each project.
- Supported airworthiness and maintenance documentation tracking for AFRC aircraft, managing configuration records and coordinating with aircraft lead operations engineers to ensure accurate aircraft configuration management.

Computational Fluid Dynamics Research Jan 2026 – Jul 2026
Georgia Institute of Technology

- Conducted **CFD simulations** in **ANSYS Fluent** across **7 flow configurations**: symmetric airfoil aerodynamics, converging-diverging nozzle internal flow, flat plate boundary layer development, transonic flow over a 3D wing, supersonic wedge flow, vertical turbine analysis, and arterial flow with and without plaque occlusion.
- Generated structured and unstructured meshes, performed **mesh refinement studies**, and verified solution convergence to ensure numerical validity across all cases.
- Quantified **pressure distributions**, **velocity fields**, and **force coefficients**; compiled comparative tabulated analyses and technical presentations for each configuration.

Leadership

Student Mentor Nov 2025 – Present
Aero Maker Space, Georgia Institute of Technology

- Mentored **200+ students** across a high-traffic makerspace with **20+ machines**, providing hands-on training and troubleshooting support across additive manufacturing (FDM, SLA, SLS), laser cutting, CNC milling, and mechanical assembly workflows.
- Performed **equipment maintenance and repair** on fabrication machinery, including **full mechanical restoration** of a band saw and ongoing repair of a table saw, diagnosing root causes and returning equipment to reliable operation.
- Managed **3D printer slicing**, job queuing, and printer maintenance to sustain high-throughput rapid prototyping operations for student projects across the facility.

Skills

Engineering: Aerodynamics, CFD Analysis, Structural Analysis, GD&T, Test Operations, Mechanical Assembly, Controls Analysis, 3D Modeling, Project Management, Aircraft Management, Propulsion System Analysis
Software: ANSYS Fluent, MATLAB, SolidWorks, Fusion 360, Creo, Inventor, Java, Javascript
Manufacturing: Additive Manufacturing (FDM, SLA, SLS), Laser Cutting, Hand Tools, Woodworking, Equipment Maintenance & Repair